

INetRadio - The Internet Radio Player

This program was made possible due to the excellent capabilities of the QB64-PE programming language and by utilizing the GuiTools Framework's UI and multi-window functionality.

Windows can be opened/closed independently. When closing the main program all other secondary windows will automatically close too. You may arrange the windows to your desire and they will remember its positions.

INetRadio is mostly self-explaining, so this is not a comprehensive documentation. The following pages just give a short overview of all windows and its controls. Most controls also have tooltips attached to it, which pop up after a second when resting the mouse over it.

INetRadio also supports UTF-8 and is using the huge Pan Unicode font arialuni.ttf to display Station names and song titles of many countries which doesn't use latin script.

Compiling and EXE

As all GuiTools Framework based applications it is for Windows only. This download contains the compiled 64-bit EXE along with the required assets (built with [QB64-PE v4.6.0](#)). If you don't trust precompiled EXEs, then simply do it yourself. The source code for this program is available as part of the latest [GuiTools Framework](#) release.

“ **Note:** Due to the UTF-8 support at least QB64-PE v3.14.0 is required to compile INetRadio. It's recommended to enable C++ optimization in the *Oprions > Compiler settings* dialog before compiling or give the flag `-f:OptimizeCppProgram=true` when compiling from command line.

You should keep INetRadio in its source folder, so it can find its INR-Assets folder. Or expressed different, when you move the EXE to your desired place, then also make sure to place a copy of the INR-Assets folder in the same location.

The INetRadio Main Window

This is the control center of INetRadio, it shows informations about the currently playing Radio Station and via the tool bar you can control the player and open other windows.



Fields

Name	Contents
Radio Station	Shows the currently selected Radio Station name.
Feeds display	Shows the incoming Feeds such as song titles or advertisements.

Controls

The **Volume** slider should be obvious, the remaining buttons work as follows:

Symbol	Function	Symbol	Function	Symbol	Function
	Stop playing and close all windows.		Open the Program Settings window.		Open the Server Informations window.
	Start playing the current Station.		Stop playing, enter standby mode.		Open the Recent Titles window.
	Select previous Station in the list.		Select next Station in the list.		Open the Stations Manager window.

Spectroscopes

The Main Window also features two frequency spectroscopes for the left and right channels respectively, each consists of 32 independent bars representing the different frequency bands. The bass frequencies are in the center and the mids and highs walk to the outside, hence a mirrored scale for the left against the right channel.

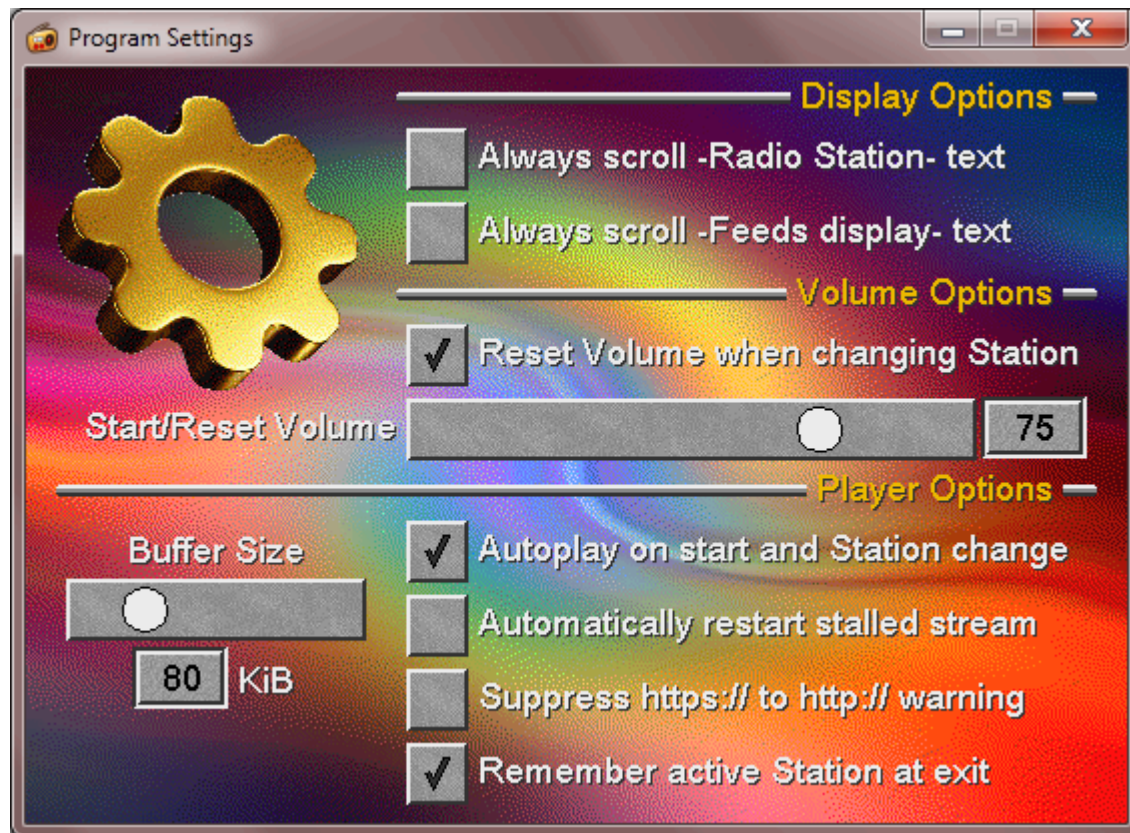
The scale is kinda logarithmic, although not an exact logarithmic function. I've rather scattered the frequencies to get a nice visual impression over the whole range with the best resolution for the bass and low mids. As higher the frequencies get as bigger is the represented bandwidth per bar, see the table below.

ID	Bars	Frequency Bandwidth (approximate)
1 - 8	8	40 - 400 Hz
9 - 16	8	400 - 1150 Hz
17 - 20	4	1150 - 1900 Hz
21 - 24	4	1900 - 3400 Hz
25 - 26	2	3400 - 4900 Hz
27 - 28	2	4900 - 7900 Hz
29 - 30	2	7900 - 13900 Hz
31	1	13900 - 19950 Hz
32	1	19950 - 24000 Hz (inaudible)

The most motion you'll see in bars 1 - 28 (40 - 7900 Hz), that's the range where normal music and singing/voice is in, while the bars 29 - 31 (7900 - 19950 Hz) mainly collect the hissing sounds of cymbal hits. The last bar 32 (19950 - 24000 Hz) is inaudible for the human ear, if you spot any motion here at all, then its almost certainly aliasing distorsion caused by resampling, which we don't wanna hear anyways.

The Program Settings Window

Here you find a couple of things that you may change to your desire. The options are organized into three semantic groups.



Controls

Group	Contents
Display Options	These options will affect the text fields in the Main Window.
Volume Options	Set start Volume and behavior when switching between Stations.
Player Options	Here you adjust all the player related values and behavior.

The Server Informations Window

Gives some additional informations returned by the server such as the Stations website or the played music genre. Note that all these informations are optional, some servers don't provide it.



Fields

Name	Contents
Name	The internal Station name used by the server, may be different from the name used in INetRadio its Stations list. Sometimes it's also misused for a stream description.
URL	The website of the Radio Station. Note this is not the URL of the actual music stream.
Genre	A list of music styles played by this Radio Station, such as Pop, Rock, Jazz etc..
Description	A more detailed description of the music stream. Some servers leave this empty and return the description in the Name tag instead.
Redirects	Shows the number of redirects followed to reach this music stream.

Controls

Symbol	Function	Symbol	Function
	Copy the website URL to Clipboard.		Open the website URL in your standard Browser.

The Recent Titles Window

This simply shows a list of the recently played song titles and Station switches.



Controls

Symbol	Function	Symbol	Function
	Mark or unmark the top entry (currently playing title) in the list. This is intended to remember titles you wanna lookup later on YouTube or iTunes.		Save the current list to disk. INetRadio will also automatically ask at exit, if marked titles exist in the list.

The Stations Manager Window

It shows a list of all preset Radio Stations in INetRadio and offers related functions to add, delete and edit Stations.

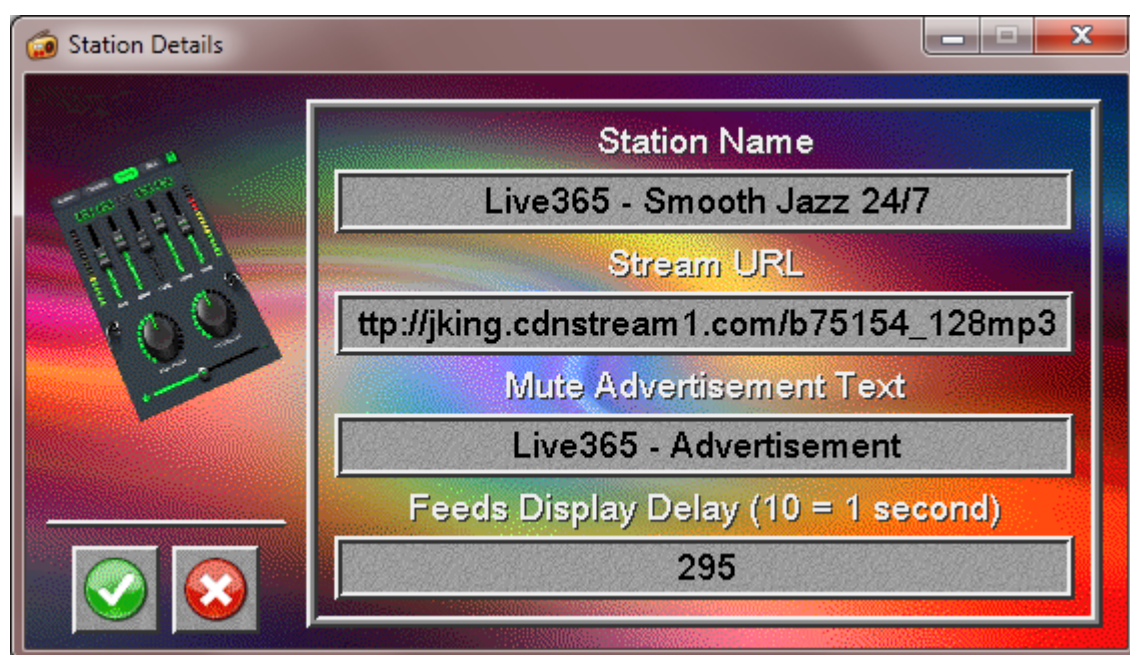


Controls

Symbol	Function	Symbol	Function
	Delete the selected Radio Station. The list can't get empty, hence you can not delete the last remaining Station.		Directly switch to the selected Radio Station. If autoplay is on it starts playing immediately.
	Add or import new Radio Stations. Supported import formats are .m3u, .m3u8 and .pls files.		Edit the details of the selected Radio Station. Opens the Station Details window.

The Station Details Window

Here you can lookup and edit the details of a Radio Station.



Fields

Note: The input fields here are the only thing not supporting UTF-8 as QB64-PE currently has no reliable unicode input methods.

Name	Contents
Station Name	The Station name used inside INetRadio, this must not necessarily match the official name.
Stream URL	The URL of the actual music stream. You may copy'n'paste it from the Radio Station's website.
Mute Advertisement Text	If you wanna auto-mute advertisements, then enter the Station's usual Feeds text shown during advertisements here. Also requires a properly set delay.
Feeds Display Delay	On some Stations Feeds come in early, set this delay so that a new song title appears right in the moment when the new song begins to play.

Controls

Symbol	Function	Symbol	Function
	Confirm your inputs and close window.		Discard all changes and close window.

My internet data is limited, how much will Web-Radio need?

Well, that depends mainly on the bitrate of the stream you're listening to, as higher the bitrate, as more data will flow through your internet line. The most preset Stations in INetRadio are 128Kbit streams, except of the "Paradise Radio" streams, which are 192Kbit. The 128Kbit streams roughly eat 57MB per hour, hence you'd hit 1GB after approx. 18 hours listening. The 192Kbit streams eat 85MB per hour, hence 1GB after approx. 12 hours.

If you have streams with other bitrates you may use the formula below:

$$\text{MB per hour} = (((\text{Kbits} * 1024) / 8) * 3600) / 1048576$$

How can I add more Radio Stations?

The Radio Station must provide the real audio stream addresses for use in external players on their website. This can either be as text which you can select and copy or as click link/button to download a .m3u, .m3u8 or .pls file that contains the stream address. What I mean is something like the **Play in your MP3 Player** button on <http://www.absolutechillout.com/> or the various links and addresses on <https://www.radioswissjazz.ch/en/reception/internet>. If the Station doesn't offer such links but rather want you to subscribe or download and use their own App, then you can assume they most probably won't even allow external players accessing their streams anyways. Note also that Stations may restrict streaming in/for/to certain countries, a quick test using the embedded player on their website will show you.

Clicking the stream address link/button will usually start downloading a single entry .m3u, .m3u8 or .pls file. You can then import the downloaded file into INetRadio in the **Stations Manager Window**. Note that plain .m3u(8) and .pls files do not contain the Station's name, instead the filename will be used as Station name and you may edit it afterwards in the **Stations Details Window**. Files using the extended .m3u(8) format contain the correct Station name. Also sometimes a downloaded file is not single entry, but contains a whole bunch of stream addresses which are all added when importing. That are usually alternative servers. If one server is down or idle, you may try another. However, I never experienced a need for that and I usually keep only the first entry and delete all other. It's up to you, if you keep or delete it.

If the Station only offers a stream address to copy, but no download, then you simply choose **Add** a new Station instead of **Import** and then copy'n'paste the address into the **Stream URL** field.

If the Station offers multiple stream encodings, then make sure to choose MP3.

Epilog

To be honest, it's very hard to find good streaming services which don't require you to subscribe or downloading and using their own App and which are nevertheless free for use and don't pack their streams with tons of commercials.

But I've found two California based services free for use (well they beg for donations) and absolutely 100% commercial free. Just the latter fact justifies a donation I guess.

<https://radioparadise.com/> (Eureka,CA)

<https://soma.fm/> (San Francisco,CA)

I've already added a couple of their channels to the preset Stations list, but there are plenty more channels to explore on their websites. The above mentioned .m3u links are at the very bottom of each channels page on **SomaFM**, simply choose any available MP3. On **Radio Paradise** use the menu "Listen >> Stream links".